Master Formula Reference Sheet

A compact, no-explanation reference to every formula in the course. Use the individual unit "Toolbox" sections if you need the plain-language explanation behind any of these — this sheet is just for fast lookup.

Limits and Continuity

$$\lim_{x\to a} f(x) = L$$

Limit laws: sums, products, quotients, and constant multiples of limits behave exactly like the corresponding operations on the individual limits (as long as the pieces exist).

Continuity at $x = a$: all three must hold:

$$f(a) \text{ is defined} \qquad \lim_{x\to a} f(x) \text{ exists} \qquad \lim_{x\to a} f(x) = f(a)$$

Limits at infinity (rational functions): compare degrees of numerator and denominator — top smaller $\to$ limit is 0; top equal $\to$ limit is the ratio of leading coefficients; top bigger $\to$ limit is $\pm\infty$.

Derivative Definition

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} \qquad f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

Basic Derivative Rules

$$\frac{d}{dx}[x^n] = nx^{n-1} \qquad \frac{d}{dx}[c] = 0 \qquad \frac{d}{dx}[cf(x)] = cf'(x)$$

$$\frac{d}{dx}[f \pm g] = f' \pm g'$$

Product Rule: $\dfrac{d}{dx}[fg] = f'g + fg'$

Quotient Rule: $\dfrac{d}{dx}\left[\dfrac{f}{g}\right] = \dfrac{f'g - fg'}{g^2}$

Chain Rule: $\dfrac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

Trig Derivatives

$$\frac{d}{dx}[\sin x] = \cos x \qquad \frac{d}{dx}[\cos x] = -\sin x \qquad \frac{d}{dx}[\tan x] = \sec^2 x$$

$$\frac{d}{dx}[\cot x] = -\csc^2 x \qquad \frac{d}{dx}[\sec x] = \sec x \tan x \qquad \frac{d}{dx}[\csc x] = -\csc x \cot x$$

Exponential and Log Derivatives

$$\frac{d}{dx}[e^x] = e^x \qquad \frac{d}{dx}[\ln x] = \frac{1}{x}$$

Inverse Trig Derivatives

$$\frac{d}{dx}[\arcsin x] = \frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}[\arccos x] = \frac{-1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}[\arctan x] = \frac{1}{1+x^2}$$

Inverse function derivative: $(f^{-1})'(a) = \dfrac{1}{f'(f^{-1}(a))}$

Motion

$$v(t) = s'(t) \qquad a(t) = v'(t) = s''(t) \qquad \text{speed} = |v(t)|$$

Speeding up: v and a same sign. Slowing down: v and a opposite signs.

Linearization

$$L(x) = f(a) + f'(a)(x-a) \qquad dy = f'(x)\,dx$$

Key Theorems

Mean Value Theorem: if f continuous on $[a,b]$, differentiable on (a,b):

$$f'(c) = \frac{f(b)-f(a)}{b-a} \text{ for some } c \text{ in } (a,b)$$

First Derivative Test: f' changes $+ \to -$: local max. f' changes $- \to +$: local min.

Second Derivative Test: $f''(c) > 0$: local min. $f''(c) < 0$: local max. $f''(c) = 0$: inconclusive.

Concavity: $f'' > 0$: concave up. $f'' < 0$: concave down. Inflection point: concavity changes.

Newton's Method: $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$

Basic Antiderivatives

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \ (n \neq -1) \qquad \int \frac{1}{x}\,dx = \ln|x| + C \qquad \int e^x\,dx = e^x + C$$

$$\int \sin x\,dx = -\cos x + C \qquad \int \cos x\,dx = \sin x + C \qquad \int \sec^2 x\,dx = \tan x + C$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\left(\frac{x}{a}\right) + C \qquad \int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$$

The Definite Integral

$$\int_a^a f(x)\,dx = 0 \qquad \int_a^b f\,dx = -\int_b^a f\,dx \qquad \int_a^b f + \int_b^c f = \int_a^c f$$

Max-Min Inequality: if $m \leq f(x) \leq M$ on $[a,b]$: $\quad m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a)$

Odd/Even symmetry (on $[-a,a]$): odd function $\to$ integral is 0. Even function $\to \int_{-a}^a f = 2\int_0^a f$.

Fundamental Theorem of Calculus

Part 1: $\dfrac{d}{dx}\left[\int_a^x f(t)\,dt\right] = f(x)$

Part 2: $\displaystyle\int_a^b f(x)\,dx = F(b) - F(a)$, where $F' = f$

Integration Techniques

u-substitution: $\displaystyle\int f(g(x))g'(x)\,dx = \int f(u)\,du$

Integration by Parts: $\displaystyle\int u\,dv = uv - \int v\,du$

Trig identities for integration:

$$\sin^2 x = \frac{1-\cos 2x}{2} \qquad \cos^2 x = \frac{1+\cos 2x}{2}$$

Trig substitution patterns:

Expression	Substitute
$\sqrt{a^2-x^2}$	$x = a\sin\theta$
$\sqrt{a^2+x^2}$	$x = a\tan\theta$
$\sqrt{x^2-a^2}$	$x = a\sec\theta$

$$\int \sec\theta\,d\theta = \ln|\sec\theta + \tan\theta| + C$$

Partial fractions: $\dfrac{P(x)}{(x-a)(x-b)} = \dfrac{A}{x-a} + \dfrac{B}{x-b}$; for a repeated factor $(x-a)^2$, include both $\dfrac{A}{x-a}$ and $\dfrac{B}{(x-a)^2}$.

Improper Integrals

$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx$$

For a discontinuity at $x = a$: $\displaystyle\int_a^b f(x)\,dx = \lim_{t\to a^+} \int_t^b f(x)\,dx$

p-integral reference: $\displaystyle\int_1^\infty \frac{dx}{x^p}$ converges iff $p > 1$. $\displaystyle\int_0^1 \frac{dx}{x^p}$ converges iff $p < 1$.

L'Hôpital's Rule

If $\displaystyle\lim_{x\to a} \frac{f(x)}{g(x)}$ is $\dfrac{0}{0}$ or $\dfrac{\infty}{\infty}$:

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

Exponential Change

$$\frac{dy}{dt} = ky \implies y = y_0 e^{kt}$$

Newton's Law of Cooling: $T(t) = T_s + (T_0 - T_s)e^{-kt}$